# Iodine and thyroid cancer risk among women in a multiethnic population: the Bay Area Thyroid Cancer Study.

Research on the relationship between iodine exposure and thyroid cancer risk is limited, and the findings are inconclusive. In most studies, fish/shellfish consumption has been used as a proxy measure of iodine exposure. The present study extends this research by quantifying dietary iodine exposure as well as incorporating a biomarker of long-term (1 year) exposure, i.e., from toenail clippings. This study is conducted in a multiethnic population with a wide variation in thyroid cancer incidence rates and substantial diversity in exposure. Women, ages 20-74, residing in the San Francisco Bay Area and diagnosed with thyroid cancer between 1995 and 1998 (1992-1998 for Asian women) were compared with women selected from the general population via random digit dialing. Interviews were conducted in six languages with 608 cases and 558 controls. The established risk factors for thyroid cancer were found to increase risk in this population: radiation to the head/neck [odds ratio (OR), 2.3; 95% confidence interval (CI), 0.97-5.5]; history of goiter/nodules (OR, 3.7; 95% CI, 2.5-5.6); and a family history of proliferative thyroid disease (OR, 2.5; 95% CI, 1.6-3.8). Contrary to our hypothesis, increased dietary iodine, most likely related to the use of multivitamin pills, was associated with a reduced risk of papillary thyroid cancer. This risk reduction was observed in "low-risk" women (i.e., women without any of the three established risk factors noted above; OR, 0.53; 95% CI, 0.33-0.85) but not in "high-risk" women, among whom a slight elevation in risk was seen (OR, 1.4; 95% CI, 0.56-3.4). However, no association with risk was observed in either group when the biomarker of exposure was evaluated. In addition, no ethnic differences in risk were observed. The authors conclude that iodine exposure appears to have, at most, a weak effect on the risk of papillary thyroid cancer.